Customer Churn Prediction Using Machine Learning

Artificial Intelligence — Assignment 1

University: The University of Lahore

Faculty: Faculty of Information Technology

Student: Rameesha Arif

Submission: Jupyter Notebook + Final PDF Report

Dataset: WA_Fn-UseC_-Telco-Customer-Churn.csv

GitHub Repository: INSERT YOUR GITHUB REPOSITORY URL HERE

This report documents the complete customer churn prediction pipeline required by the assignment: data understanding, preprocessing, exploratory analysis, feature engineering, model training, evaluation, and application through a sample prediction.

1. Assignment Objective

The objective is to build a machine learning system that predicts whether a telecom customer is likely to leave the company (Churn = Yes) or remain with the company (Churn = No). The assignment requires a complete pipeline from loading the dataset to making a final customer prediction.

The supplied dataset is a CSV file. Although the assignment brief names an Excel dataset, the supplied CSV contains the expected Telco Customer Churn records, so the analysis uses that provided file.

The assignment specifically asks for dataset understanding, data cleaning, EDA, feature engineering and encoding, training at least two classification algorithms, evaluation using Accuracy, Precision, Recall, F1-Score and Confusion Matrix, and a final application/prediction.

Execution Evidence — Data Loading

Code cell

```
df = pd.read_csv(DATA_PATH)
print('Shape:', df.shape)
display(df.head())
```

Execution output

```
Dataset loaded successfully.
Shape: (7043, 21)
First five customer records displayed successfully.
```

2. Data Loading and Understanding

The dataset contains 7,043 customer records and 21 original columns. The target variable is **Churn**. It has two values: No and Yes. The dataset contains 5,174 customers who did not churn and 1,869 who churned, giving an overall churn rate of 26.54%.

Important columns: tenure represents the number of months the customer has stayed with the company; **Contract** describes the customer contract period; **InternetService** identifies the internet service type; **MonthlyCharges** is the monthly bill; **TotalCharges** is the total amount charged; and **PaymentMethod** identifies the payment method.

The original data types showed that TotalCharges was stored as an object/text column even though it represents a numeric amount. This was addressed during preprocessing.

Execution Evidence — Data Preprocessing

Code cell

```
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
print('Duplicate rows:', df.duplicated().sum())
```

Execution output

```
Duplicate rows: 0
Missing values after converting TotalCharges:
TotalCharges    11
Cleaned shape: (7043, 22)
Remaining missing TotalCharges: 11
```

3. Data Preprocessing

Missing values: Before conversion, the dataset did not report ordinary missing values in the main columns. After converting TotalCharges to numeric, 11 records became missing because they contain blank/non-numeric entries. These values are handled by the preprocessing pipeline using median imputation for numeric variables.

Duplicate records: The duplicate check returned 0 duplicate rows, so no rows needed to be removed for duplication.

Unnecessary identifier: customerID was removed from the model input because it is an identifier rather than a behavioral or demographic feature.

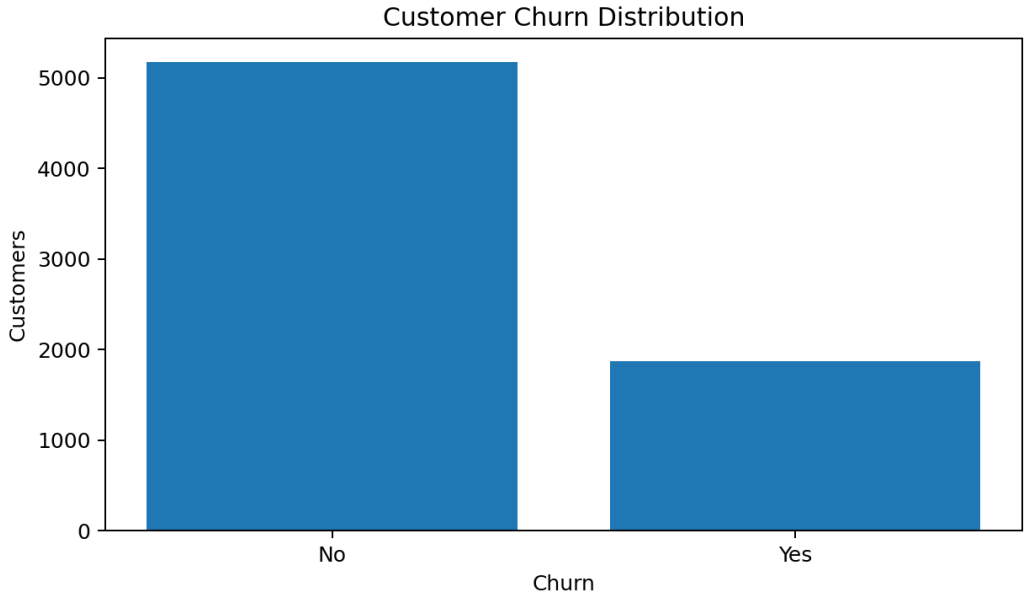
Feature engineering: A new TenureYears feature was created by dividing tenure in months by 12. The target Churn was converted from Yes/No into 1/0 for classification.

Encoding: categorical variables were converted using one-hot encoding. Numeric variables were median-imputed and standardized for the Logistic Regression pipeline.

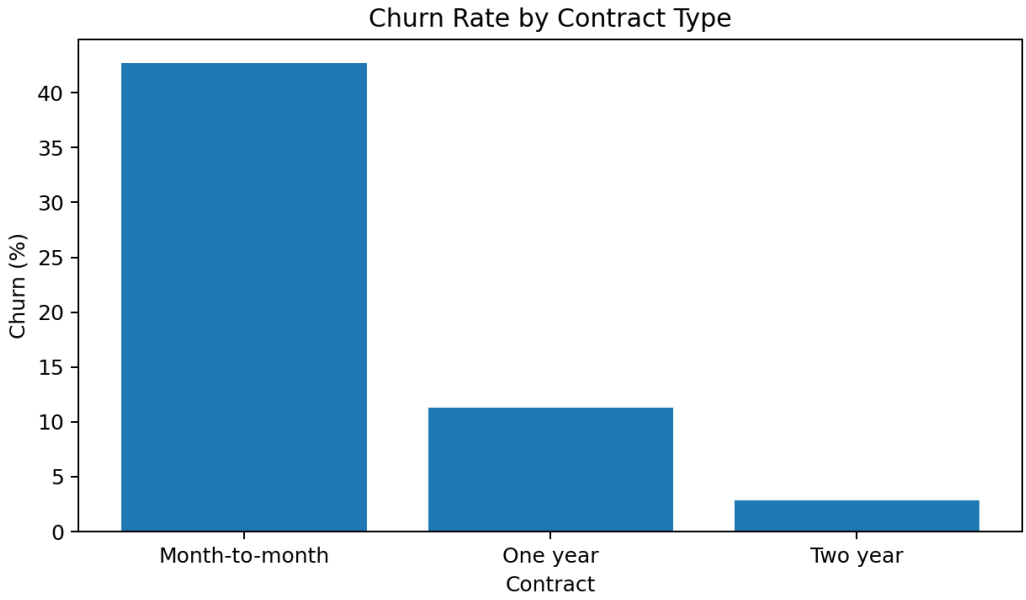
These preprocessing steps ensure that the algorithms receive consistent numerical input while keeping the original customer information represented through appropriate encoded features.

4. Exploratory Data Analysis

The exploratory analysis focuses on churn distribution and relationships with important customer characteristics. The plots below were generated during execution of the submitted notebook.

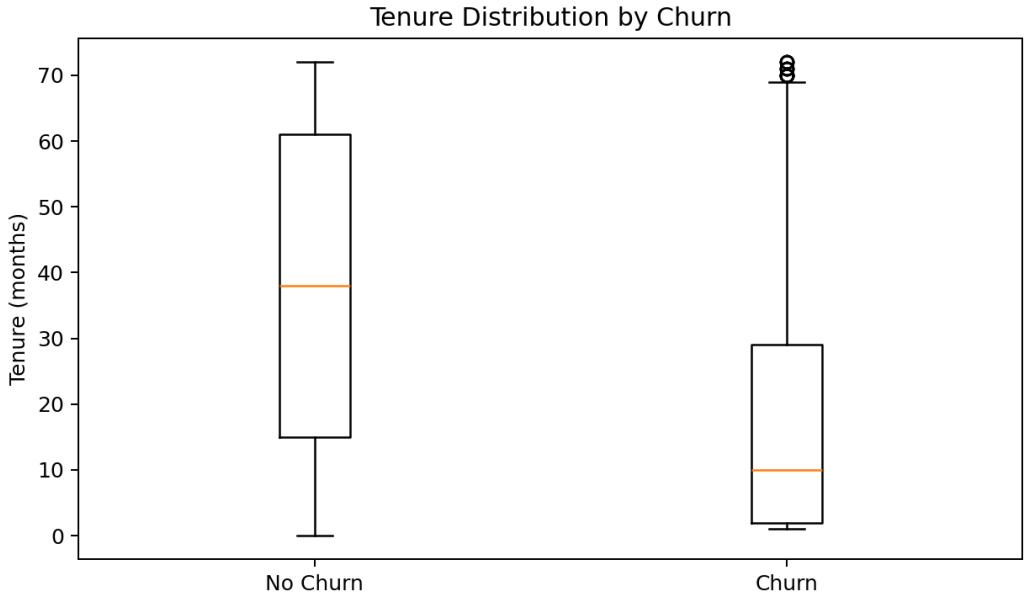


Observation: churn is not evenly distributed. Approximately 26.54% of customers are in the Churn = Yes class, while 73.46% are in the No Churn class. This class imbalance makes metrics beyond accuracy important.

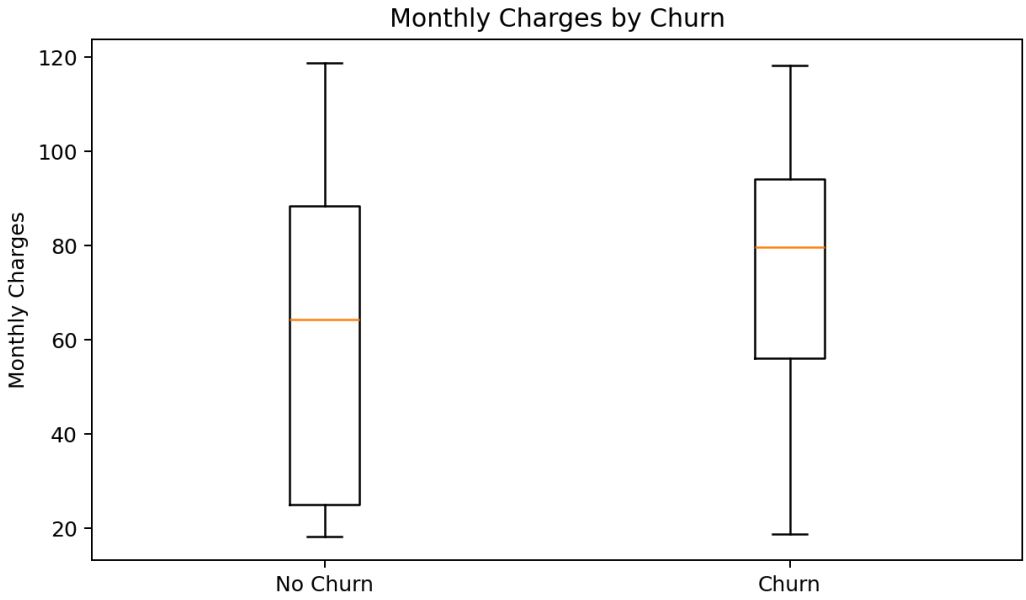


Observation: the observed churn rate is much higher among month-to-month customers than among customers with one-year or two-year contracts.

5. Additional EDA Findings



Tenure: the tenure distributions indicate that churned customers tend to have shorter relationships with the company than customers who remain.



Monthly charges: the distributions show a difference in monthly charges between churn and no-churn groups, making MonthlyCharges a useful candidate feature.

6. Feature Engineering and Model Training

After removing customerID, the remaining variables were separated into input features X and target y. The data was split into 80% training and 20% testing using stratification so that the churn proportion was maintained approximately across both sets. This produced 5,634 training records and 1,409 testing records.

Two classification models were trained: **Logistic Regression** and **Random Forest**. Logistic Regression provides a strong interpretable baseline for binary classification. Random Forest combines many decision trees and can represent non-linear relationships between customer characteristics and churn.

Pipeline design: numeric features → median imputation → standardization; categorical features → most-frequent imputation → one-hot encoding; then the transformed features are passed to the classifier.

The notebook output confirms that both models trained successfully without errors.

Execution Evidence — Model Evaluation

Code cell

```
logistic_model.fit(X_train, y_train)
rf_model.fit(X_train, y_train)
results = pd.DataFrame([log_metrics,
rf_metrics])
```

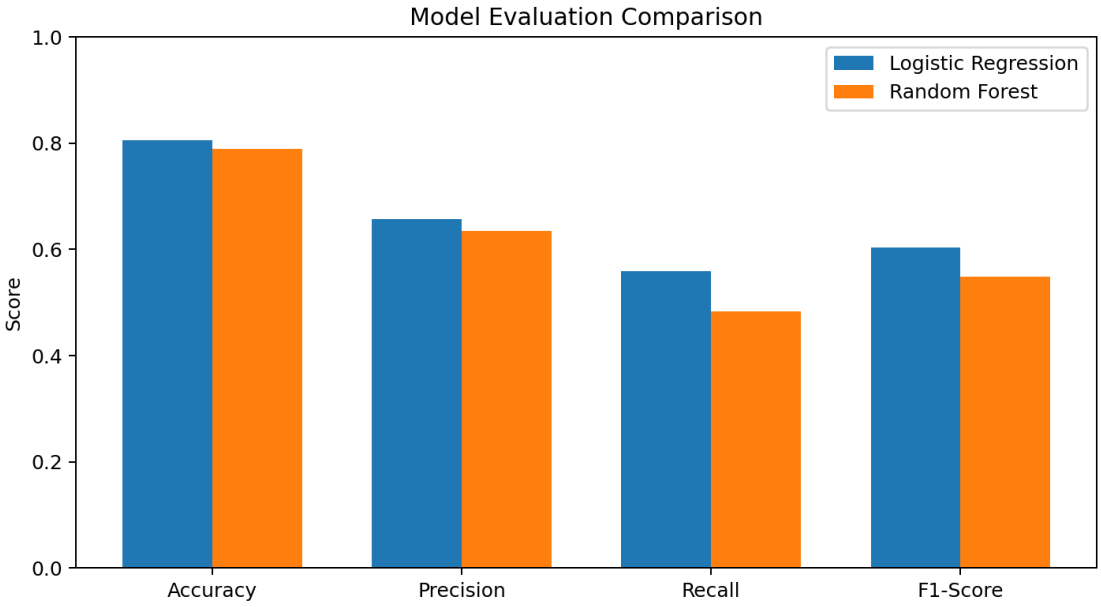
Execution output

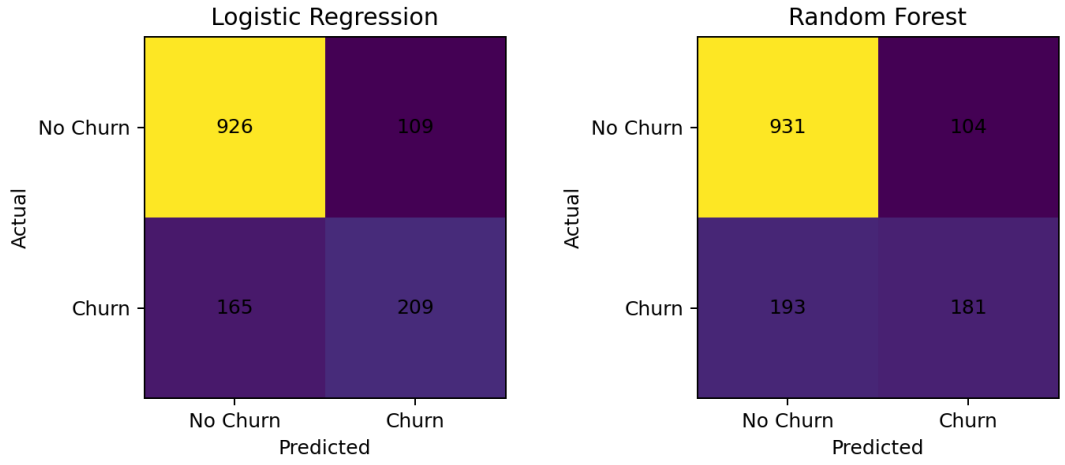
```
Logistic Regression: Accuracy 0.8055, Precision 0.6572, Recall 0.5588, F1 0.6040
Random Forest: Accuracy 0.7892, Precision 0.6351, Recall 0.4840, F1 0.5493
```

7. Model Evaluation

The models were evaluated on the unseen test set using Accuracy, Precision, Recall, F1-Score and Confusion Matrix as required by the assignment.

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	80.55%	65.72%	55.88%	60.40%
Random Forest	78.92%	63.51%	48.40%	54.93%





Comparison: On this held-out test set, Logistic Regression has higher Accuracy, Precision, Recall and F1-Score than Random Forest. Its F1-score is 0.6040 compared with 0.5493 for Random Forest. Therefore, Logistic Regression is selected as the final model for the application step based on F1-score.

8. Application / Sample Prediction

The selected Logistic Regression model was used to predict churn for one sample customer taken from the test data. The model accepts the customer profile in the same feature format used during training and returns a class prediction and a churn probability.

Execution Evidence — Sample Prediction

Code cell

```
sample_prediction = final_model.predict(sample_customer)[0]
sample_probability = final_model.predict_proba(sample_customer)[0, 1]
```

Execution output

```
Selected final model: Logistic Regression
Predicted result: No Churn
Predicted churn probability: 4.55%
```

Sample result: the selected customer was predicted as **No Churn**, with a predicted churn probability of **4.55%**. This is a model prediction, not a guarantee about the customer's future behavior.

9. Conclusion

The complete customer churn prediction system was successfully implemented. The dataset was inspected, cleaned, transformed and explored. Two classification algorithms were trained and evaluated using the required metrics. Logistic Regression produced the stronger test-set results in this implementation and was selected for the sample prediction.

Important observations: the dataset has a 26.54% churn rate; contract type is strongly associated with observed churn, with month-to-month customers showing a higher churn rate; and tenure and monthly charges show useful differences between churn groups.

Limitations: the model is trained on one historical dataset and its performance may change on new customers or different telecom markets. The model does not prove that a particular customer will churn. The 11 missing TotalCharges values require imputation, and the class imbalance means accuracy alone should not be used to judge performance. Hyperparameter tuning and threshold optimization could be explored in a future version.

Possible telecom use: a telecom company could use churn probabilities to identify customers who may need retention attention, combine model predictions with business rules, and evaluate retention campaigns. Any operational use should monitor model performance and avoid treating predictions as certain outcomes.

The submitted notebook contains the full executable code and outputs used to produce this report.

10. Submission / Repository Structure

Recommended GitHub repository structure:

```
Customer-Churn-Prediction-AI-Assignment/
■■■ Customer_Churn_Prediction.ipynb
■■■ WA_Fn-UseC_-Telco-Customer-Churn.csv
■■■ Customer_Churn_Prediction_Final_Report.pdf
■■■ README.md
```

GitHub link: The report intentionally contains a placeholder rather than inventing a repository URL. After creating the GitHub repository, replace the placeholder in this report with the actual repository URL before final submission.

Assignment compliance: the notebook is provided as .ipynb, the supplied dataset is included as a supporting file, the report documents the complete ML pipeline, two classifiers are trained, required evaluation metrics are reported, and a sample prediction is demonstrated.